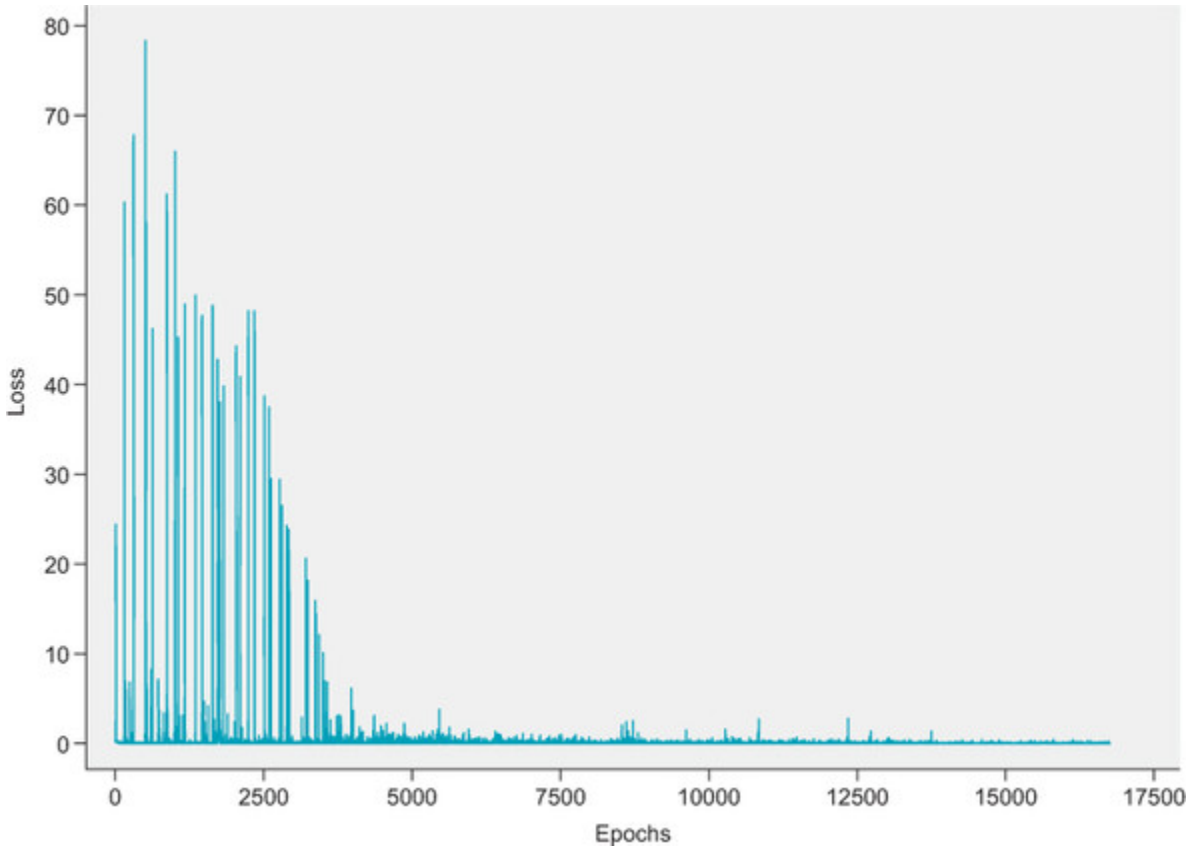


The loss plot is pretty noisy, but the moving average of the plot is significantly trending toward zero. This gives us some confidence the training worked, but we'll never know until we test it. We've written up a simple function in listing 3.4 that allows us to test the model on a single game.

Listing 3.4. Testing the Q-network

```
def test_model(model, mode='static', display=True):
    i = 0
    test_game = Gridworld(mode=mode)
    state_ = test_game.board.render_np().reshape(1,64) +
np.random.rand(1,64)/10.0
    state = torch.from_numpy(state_).float()
    if display:
        print("Initial State:")
        print(test_game.display())
    status = 1
    while(status == 1):                    1
        qval = model(state)
        qval_ = qval.data.numpy()
        action_ = np.argmax(qval_)         2
```